

Validation of a Mobile-Based Semen Analysis Application (MAKSA) Against CASA and Subjective Method for Post-Thaw Evaluation of Bovine Semen

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Abstract

Accurate assessment of post-thaw semen quality is a prerequisite for the success of artificial insemination programs in bovines. Sperm motility is one of the most informative indicators of sperm viability following cryopreservation. Computer-assisted semen analysis (CASA) is widely accepted as a reference method for objective motility evaluation; however, its routine use is constrained by high cost, limited portability, and laboratory dependency. Mobile-based semen analysis applications have recently been introduced as potential low-cost alternatives, but independent validation data for post-thaw bovine semen remain limited.

The present study was conducted to validate the performance of a mobile-based semen analysis application (MAKSA) by comparing its post-thaw semen motility and concentration outputs with those obtained using CASA and conventional manual microscopic assessment. Frozen–thawed semen samples from Sahiwal cattle bulls and buffalo bulls were evaluated under standardized laboratory conditions using the same phase-contrast microscope. Average values of total motility, progressive motility, and sperm concentration were compared across methods. The results demonstrated that MAKSA produced motility estimates that followed trends similar to CASA, whereas sperm concentration values showed marked differences among methods. The findings indicate that MAKSA can serve as a practical tool for post-thaw semen motility screening, although it is not suitable for precise sperm concentration estimation.

Keywords: post-thaw semen, sperm motility, CASA, MAKSA, bovine, artificial insemination

Introduction

Artificial insemination plays a central role in genetic improvement and reproductive management of bovine herds. The fertility outcome of artificial insemination programs depends largely on the quality of semen used, particularly after freezing and thawing. Cryopreservation imposes physical and biochemical stress on spermatozoa, leading to reduced motility and compromised functional competence. Therefore, reliable evaluation of post-thaw semen quality is essential before semen is approved for use.

Among the various semen quality parameters, sperm motility, especially progressive motility, has been consistently associated with fertilizing potential. Traditionally, sperm motility has been assessed manually under a phase contrast microscope. Although this approach is inexpensive and simple, it is inherently subjective and prone to inter- and intra-observer variability. To overcome these limitations, computer-assisted semen analysis systems were developed to provide objective, repeatable, and quantitative assessment of sperm movement.

Despite their advantages, CASA systems require expensive equipment, controlled laboratory conditions, and trained personnel. These constraints limit their application in routine semen screening and field-level settings, particularly in developing countries. Recently, mobile phone-based semen analysis applications have been developed with the aim of providing rapid and low-cost semen evaluation. MAKSA is one such application that estimates semen motility parameters from short video recordings captured through a microscope-mounted smartphone.

However, before such tools can be recommended for routine use, their performance must be critically evaluated against established reference methods. The present study was therefore designed to validate MAKSA for post-thaw bovine semen evaluation by directly comparing its outputs with CASA and manual microscopic assessment under standardized conditions.

Objective

To validate the performance of the mobile-based semen analysis application (MAKSA) for post-thaw bovine semen evaluation by comparing its motility and concentration estimates with those obtained using CASA and manual microscopic assessment.

Materials and Methods

Study location and semen samples

The study was conducted at the University Semen Analysis and Cryopreservation Laboratory, Department of Clinical Sciences, Faculty of Veterinary and Animal Sciences, Bahauddin Zakariya University, Multan, Pakistan. Frozen semen straws from Sahiwal cattle bulls (n=7) and buffalo bulls (n=3) were used. All semen straws were stored in liquid nitrogen (−196 °C) until evaluation.

Thawing procedure

Frozen semen straws were thawed in a water bath maintained at 36–37 °C for 30 seconds. After thawing, the straws were dried, cut, and the semen was expelled into pre-warmed tubes for immediate analysis.

Manual motility evaluation

For motility assessment, semen samples were prepared using a simple glass slide and coverslip. Slides and coverslips were pre-warmed to approximately 37 °C to minimize temperature-related motility loss. A small drop of semen was placed on the slide and covered gently to ensure uniform spreading. Manual motility assessment was performed using a phase-contrast microscope. At least five random microscopic fields were examined per sample, and a minimum of 100 spermatozoa were evaluated. Total motility and progressive motility were estimated by a single trained observer. The same slides were used for MAKSA and CASA analysis.

MAKSA analysis

For MAKSA analysis, semen samples were examined at 200× magnification using the same phase-contrast microscope. A smartphone camera was fixed to the microscope eyepiece, optical zoom was set four times, and videos of six second duration were recorded selected fields. The recorded videos were analyzed using the MAKSA application to obtain total and progressive motility and concentration estimates.

CASA analysis

CASA evaluation was performed using a laboratory-based CASA system connected to a phase-contrast microscope. The same semen samples prepared on glass slides with coverslips were analyzed, and total and progressive motility values were recorded automatically by the CASA software.

Sperm concentration assessment

Sperm concentration was determined manually using a hemocytometer following dilution (1:200) of semen in formal saline. The hemocytometer provides a defined chamber depth and grid area, allowing calculation of sperm concentration on a volumetric basis. Sperm concentration was assessed in CASA and MAKSA by default setting.

Results

Average post-thaw semen parameters obtained using manual microscopy, CASA, and MAKSA are summarized in Table 1. CASA consistently yielded the highest average values for total motility, whereas manual assessment produced moderate values with greater variability. MAKSA produced intermediate motility values that followed trends similar to those observed with CASA.

In contrast, marked differences were observed in sperm concentration estimates. Manual haemocytometer-based assessment yielded the highest average concentration values, while CASA and MAKSA produced lower estimates.

Table. Average (Mean $\pm$ SE) post-thaw semen parameters assessed by Manual microscopy, CASA, and MAKSA

Parameter	Method	Mean $\pm$ SE	Relative level
Total motility (%)	Manual	54.0 $\pm$ 10.3	Moderate
	CASA	91.0 $\pm$ 2.1	High
	MAKSA	79.0 $\pm$ 11.5	Moderate-High
Progressive motility (%)	Manual	38.0 $\pm$ 8.6	Moderate
	CASA	40.0 $\pm$ 2.5	Moderate
	MAKSA	52.0 $\pm$ 9.1	Moderate-High
Sperm concentration (million/ml)	Manual	195 $\pm$ 9.2	High
	(haemocytometer)		
	CASA	68 $\pm$ 5.3	Low-Moderate
	MAKSA	34 $\pm$ 11.7	Low

Discussion

The present study evaluated the performance of a mobile-based semen analysis application for post-thaw bovine semen assessment under standardized laboratory conditions. The results demonstrate that MAKSA can provide motility estimates that are broadly comparable in trend to those obtained using CASA, whereas sperm concentration values differ substantially among methods.

The comparability of motility results can be attributed to the fact that motility is primarily a movement-based parameter. Both CASA and MAKSA analyze sperm motion from video recordings, enabling detection of motile and immotile spermatozoa. Although MAKSA employs simpler tracking algorithms than CASA, it is still capable of capturing overall motility trends when factors such as temperature, sperm concentration, and illumination are adequately standardized. The use of the same phase-contrast

microscope for manual, CASA, and MAKSA analyses further minimized methodological bias.

In contrast, sperm concentration estimation is highly dependent on the presence of a defined counting volume. Manual haemocytometer-based assessment relies on a fixed chamber depth and known grid area, allowing accurate volumetric calculation of sperm concentration. Conversely, CASA and MAKSA concentration estimates in the present study were derived from samples prepared on simple glass slides with coverslips. Under these conditions, concentration estimates depend on the size of the semen droplet, the uniformity of sample spreading under the coverslip, and variability in chamber depth. As a result, CASA and MAKSA tended to underestimate sperm concentration relative to haemocytometer-based counts.

The effect was more pronounced in MAKSA due to additional optical limitations associated with smartphone-based imaging, including restricted field of view and sensitivity to focal depth. These findings indicate that while MAKSA is suitable for motility screening, its concentration outputs should be interpreted with caution.

Conclusion

CASA remains the most reliable and objective method for post-thaw bovine semen motility assessment. The mobile-based application MAKSA demonstrated acceptable performance for motility evaluation when standardized protocols were followed and showed better consistency than manual subjective assessment. The large variation observed in sperm concentration among methods is likely attributable to the use of simple glass slides and coverslips, where concentration estimates depend on droplet size and uniformity of sample spreading. CASA provides more accurate concentration measurements when standardized chambers such as Leja or GoldCyto slides with fixed depth are used. Similarly, MAKSA concentration estimates may improve if a uniformly spread semen droplet or standardized chamber is employed. Therefore, MAKSA may

be recommended as a practical, low-cost tool for routine post-thaw semen motility screening in situations where CASA facilities are not available.